Shreyansh Ajmera

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Structural engineer with around 7 years of experience in bridge design, from concept to detailed design of highway and infrastructure projects; looking to expand that foundation into a broader role encompassing assessment, digital delivery and smart infrastructure. Proficient in Eurocodes and Norwegian standard N400, with working knowledge of AASHTO LRFD. Experienced in FE modelling and parametric workflows using SOFiSTiK, Grasshopper, Rhino, and LUSAS, applied directly on live projects to streamline model generation and improve design efficiency. Familiar with BIM processes and coordinated model delivery, including IFC model checking and Trimble Connect. Keen to develop further in Python-based automation, advanced data processing and smart bridge technologies. Active in mentoring junior engineers and delivering internal training on structural analysis tools.

Area of Expertise

Analysis and Design of Bridges, Parametric Modelling and Optimization - Computational Design

Professional Memberships

Institution of Civil Engineers (ICE) – Graduate Member
Currently pursuing Chartered Engineer (CEng MICE) status through the ICE.
25 out of 35 attribute levels completed; aiming to achieve chartered status in 2027.
CITB CSCS Card
CSCS Card for Managers and Professionals valid till 2029

Software proficiency

Analysis platforms
SOFiSTiK, LUSAS, Karamba3D, Idea StatiCa, Oasys AdSec, ASBD

Programming
Python, Grasshopper, Excel VBA, Mathcad

Modelling & CAD
Rhino, AutoCAD, Solibri Office, Trimble Connect

Career

Design Engineer – Major Projects, Ramboll UK (June'23 – Present)
Assistant Design Engineer – Ramboll India Pvt Ltd (July'19 – June'23)

Key Project Experience

E6-Ranheim-Sveberg, Norway (May'25-Jan'26)
Client: HÆHRE
For concept design stage of the K31 Vulu Bridge, a three-span integral post-tensioned slab deck bridge with a maximum span of 33 m, developed the analytical line model in SOFiSTiK and prepared load calculation sheets. Carried out structural analysis and finalised the preliminary cross-sections of the deck, inner and outer piers, and pier heights. Optimised the post-tensioning cable profile to achieve optimal load distribution. The design and analysis were performed in accordance with Manual N400 (Norwegian standard) and Eurocodes.
For detailed design, developed a fully parametric SOFiSTiK shell model and used DECREATOR to integrate the results, used in-house advanced code blocks for actions, combinations, live load generation and design; developing, testing and debugging these blocks further. Developed code blocks for automatic generation of certain parts of the report. Coordinated with the BIM team on paperless delivery, checking the IFC model in detail, developing a detailed attribute spreadsheet to export to BIM model and checking it from geometry to reinforcement to attributes.

Ny Ölfusárbrú, Iceland (Sept'24-April'25)
Client: ÞG-Verk
Detailed design of the cable-stayed bridge abutments integrating advanced tools such as SOFiSTiK, Rhino–Grasshopper, and Excel. Extracted geometric and design data from a central SPOT (Single Point of Truth) Excel database via the Rhino–Grasshopper interface to streamline and parametrise the analytical model. Developed a parametric shell model in SOFiSTiK and completed the detailed design of the abutments. Prepared reinforcement sketches for technician input, coordinating closely with teams in India and Denmark. Collaborated with the geotechnical team to determine earth pressure coefficients

under various design scenarios and to finalise the design loads. Reviewed and commented on the 3D model using Trimble Connect to ensure coordination and accuracy.

Genova Breakwater, Italy (Sept'23-Sept'24)
Client: F&M ingegneria
Detailed design of the breakwater caissons under extreme wave conditions. Developed a parametric model in SOFiSTiK for multiple caissons with varying geometries and loading scenarios. Prepared wave pressure calculation spreadsheets based on the Coastal Engineering Manual, Part VI; created a python script for the automatic generation of stability calculations replacing variables in Mathcad and extracting the pdf and output to the design report. Coordinated with a team of technicians in India to develop detailed caisson drawings.

E6-Ranheim-Værnes, Norway (July'21-Feb'23)
Client: Acciona Construction
Detailed design of several overbridges on the E6 highway, all designed as three-span voided deck integral bridges. Involved in concept design tasks such as determining dimensions, void arrangements, pier locations and numbers, and initial tendon profiles. Prepared load calculation sheets and grillage models in SOFiSTiK and applied them accordingly. Designed longitudinal members and piers using SOFiSTiK's in-built modules. Developed local shell models for the design of ballast and wing walls. Also completed the transverse deck design, and the design of pad foundations, pile caps, and piles using Mathcad. Created project schedules and performed self–QA checks to ensure timely, high-quality deliverables. Prepared technical reports, BoQ, Safety-in-Design documentation, and checklists, and coordinated with technicians on the 3D model for 400MMI.

Laos Bridge, LAO PDR (Mar'20 -Aug'20)
Client: Lao PDR Ministry of Information, Culture and Tourism
Detailed design of a three-span simply supported beam–slab bridge using LUSAS and in-house Excel spreadsheets. Developed design tools for all bridge components in accordance with AASHTO standards. Coordinated with technicians to prepare drawings, assisted the principal engineer in compiling the design report, and produced the BoQ. Coordinated with client on the design comments, developed spreadsheet for post-tensioned beam loss calculation, tracing back the approved design decisions and responded to the comments.

Delhi Noida International Airport (Sept'20-Jan'21)
Client: Yamuna International Airport Private Limited, India
Involved in the tender design of airside and landside culverts, prepared parametric analysis and design file by linking excel and STAAD.Pro for the design of variable cell culverts. Prepared a list of drawings, reviewed specifications, collaborated with technicians on the preparation of drawings.

Assessment works (Jan'26-Present)
Client: Multiple
Currently working on multiple jobs including

- CAT2 check of the assessment on a bridge damaged due to fire using NR guidance notes CIV/025
- Strengthening of a damaged beam using CFRP using TR 55
- Optioneering for the layout of piping network crossing the bridge

Technical Initiatives & Automation Projects

In addition to project-specific tasks, I have developed Grasshopper scripts to perform linear structural analysis using Karamba3D, enabling quick parametric studies within the Grasshopper environment. I also created customised interoperability tools by linking Grasshopper with LUSAS and SOFiSTiK to streamline the generation and updating of analytical models. As a design optimisation exercise, I modelled a tied-arch bridge and used the Galapagos evolutionary solver in Grasshopper to identify optimal number of cables and span to height ratio minimising the deflections and the sum of bending moments in the tie members.

Professional Development & Training

LUSAS Software Training – Grillage Modelling & Vehicle Load Application
Internal training, 2022

Delivered a structured training session to colleagues on grillage modelling and vehicular loading in LUSAS, including the use of moving load and Vehicle Load Optimisation (VLO) features. Covered practical modelling tips and demonstrated how to interpret results for bridge design applications.

Introductory Training – SOFiSTiK and Grasshopper
Ongoing Internal Sessions
Provided informal technical training to junior staff on getting started with SOFiSTiK and Rhino-Grasshopper workflows for parametric modelling and structural analysis, illustrating concepts with real project examples.

Academic Qualification

Master of Technology in Structural Dynamics from IITR, India (2017-2019)
Bachelor of Technology in Civil Engineering from UKA Tarsadia University, India (2012-2016)

Languages Known

English – Professional Proficiency

Hindi – Mother tongue